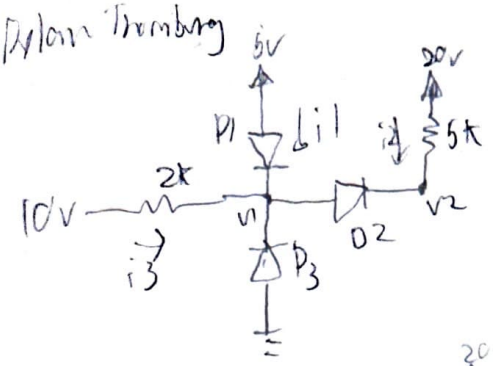
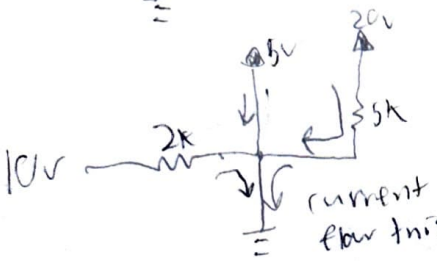
Dylan Thornburg

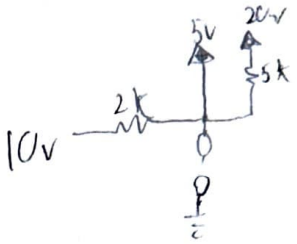
1.



It looks like all diodes will be off, but I don't want to prove it that way. Instead I will prove it by showing what happens when all/any are on



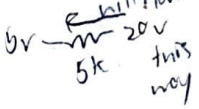
current wants to flow this way and cannot b/c voltage at D3's cath would be higher than D3's anode, D3 is off



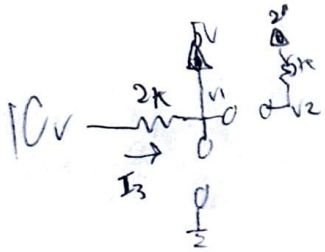
In this cas v1=5 which means

$\frac{20-5}{5k} = 3mA = i2$

can also think like this



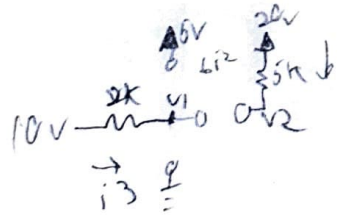
since current cannot flow that way for D2, it is off



$I_3 = \frac{10-v1}{2k}$, $v1=5$, $I_3 = 2.5mA$

Aggin we have the same issue where I_3 wants to travel incorrectly through D1.

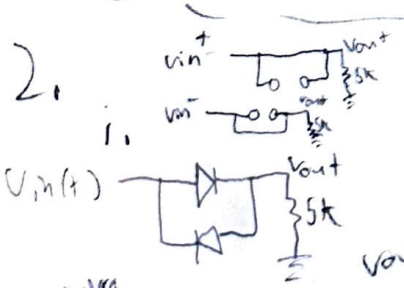
Thus D1 is off



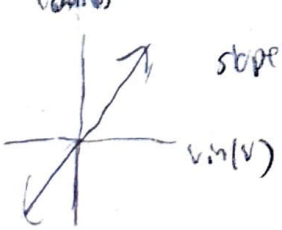
$v1 = 10V$
$v2 = 20V$
$i1 = i2 = i3 = 0A$

2.

i.



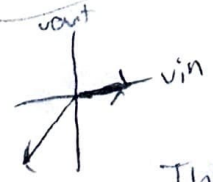
$v_{out} = v_{in}$

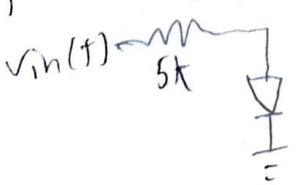


slope = 1

The two diodes act as a short

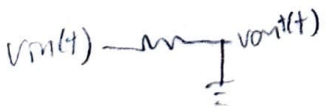
ii



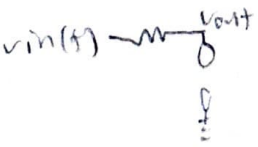


This circuit only allows negative voltages through. $v_{out} = v_{in}$ only when vin is negative. It is a Negative half wave rectifier

when vin(t) is pos

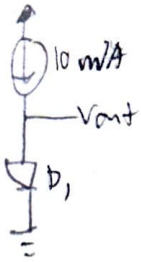


when vin(t) is neg



3.

figure 2



1.

Vout = 0

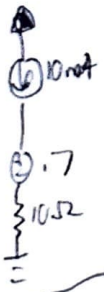
$\boxed{\text{Vout} = 0V}$

2.

Vout = .7V

$\boxed{\text{Vout} = .7V}$

3.



$.7 + (.01)(10) = .8$

$\boxed{\text{Vout} = .8V}$

4.

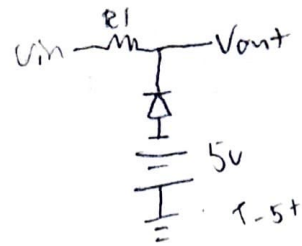
we have 10mA

so V = .6

$\boxed{\text{Vout} = .6V}$

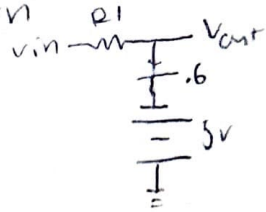
5. It looks to be about 2.1; $\boxed{\text{Vout} = 2.1\ V}$

4. N1

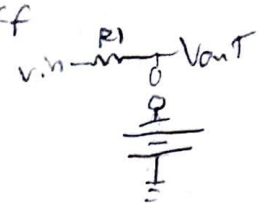


$-5 + .6$

on

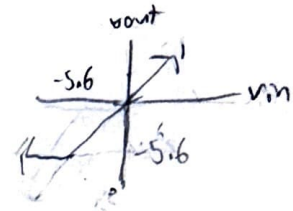


off

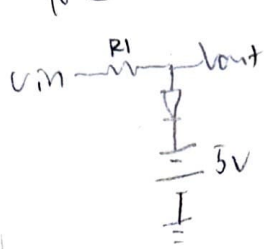


on when $vin < -5.6V$

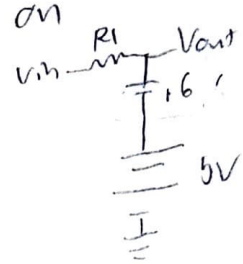
off when $vin \geq -5.6$



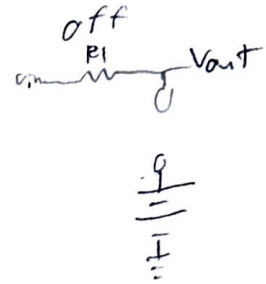
N2



on



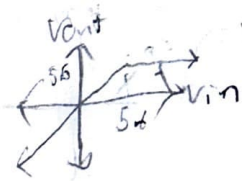
off



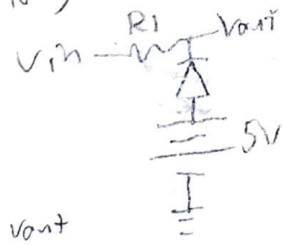
$.6 + 5 = 5.6$

on when $vin > 5.6$

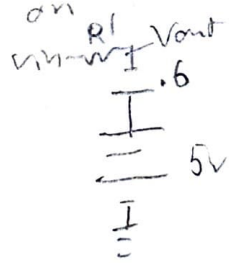
off when $vin \leq 5.6$



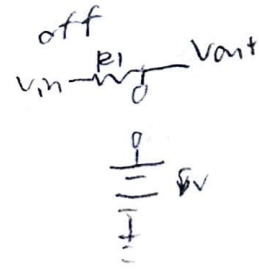
N3

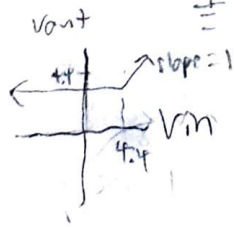


on



off



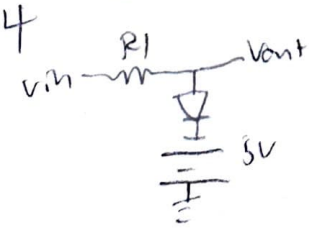


$-.6 + 5 = 4.4$

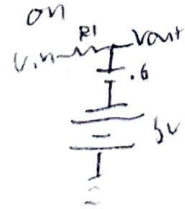
on when $vin \leq 4.4$

off when $vin \geq 4.4$

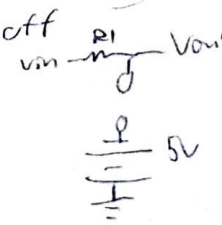
N4

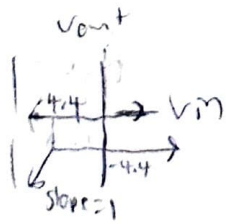


on



off





$.6 - 5 = -4.4$

on when $vin \geq -4.4$

off when $vin \leq -4.4$

4 ext

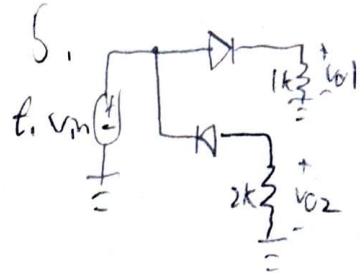
d.

Choice 2 would be best as it clamps when vin goes above 5.6 V. If it was ideal, it would clamp right at 5V. It's the closest circuit to what we want.

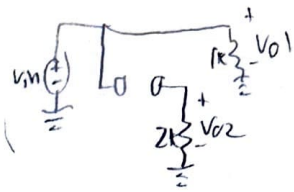
e. A voltage drop of 1V makes it worse as it'll clamp at 6V now. This will allow .4 more V then you want

5.

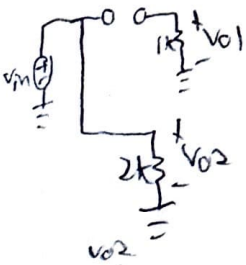
f.

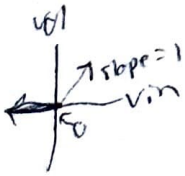


when vin is +

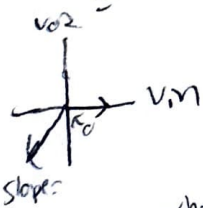


when vin is −





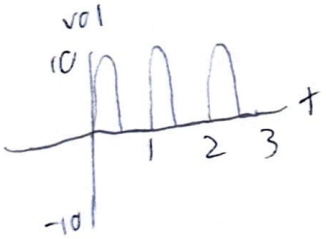
Vo1 = vin when vin > 0

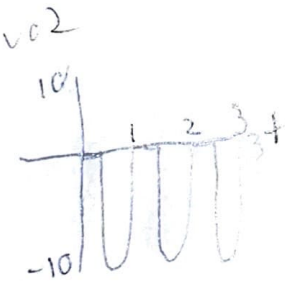


Vo2 = vin when vin < 0

5.

g. for vin = 10 sin 2πt





$$\int_0^{\frac{1}{2}} 10 \sin 2\pi t \, dt$$

$$10 \left[\frac{-\cos(2\pi t)}{2\pi} \right]_0^{\frac{1}{2}}$$

$$10\left(\frac{1}{2\pi}\right) - -10\left(\frac{1}{2\pi}\right) =$$

$$\boxed{Vo1\ avg = 3.18V}$$

$$\boxed{\text{Peak } D1 = \frac{10}{1k} = 10mA}$$

Max reverse voltage D1

−10v

$$\boxed{V_r = 10v}$$

Vo2 avg will just be Vo1 but neg.

$$\boxed{Vo2\ avg = -3.18v}$$

Peak D2 = 0 unless we count magnitude as peak in which

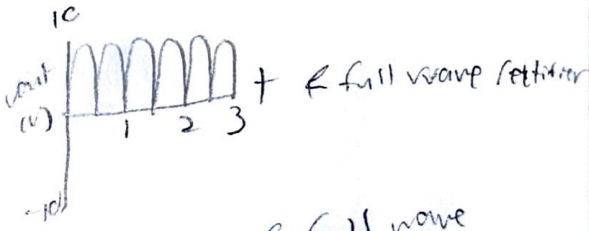
$$\boxed{D2 = \frac{-10}{2k} = -5mA \text{ or a peak of } 5mA \text{ (magnitude)}}$$

Max reverse voltage D2

10v but we abs value them so

$$\boxed{10V = V_r}$$

h.



← full wave rectifier

avg voltage of full wave rect. $= \frac{2v_m}{\pi} - V_{do}$

$$\frac{2 \cdot 10}{\pi} - 0 = 6.37$$

$$\boxed{V_{out}\ avg = 6.37V}$$

The designer made a full wave rectifier b/c we only have positive output even w/ negative input, V1−V2 only results in positive output